JEPA-x: Cross-Predictive Physics Grounding for Forecastable Latent Dynamics

Kehan Wen Ziming Li Siyuan Luo Fan Shi

Department of Electrical and Computer Engineering, National University of Singapore

NUS National University of Singapore

Human-centered Robotic Lab

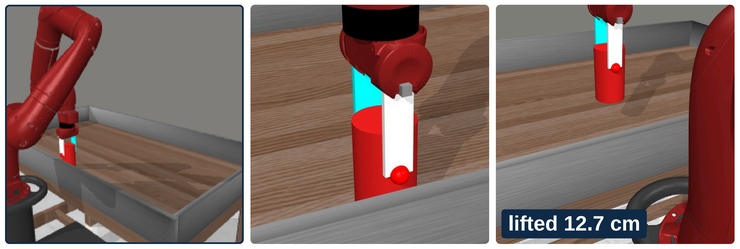
CoRL

1 Problem

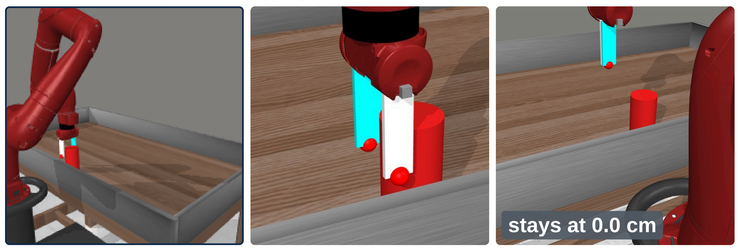
A visual world model must predict how a scene evolves from pixels alone. **Two scenes can look almost identical to the camera yet behave differently under the same action.**

Model's view 224 px input	**Actual scene** side view	**After the same lift**

Scene A · block held in the jaws



Scene B · block just behind the jaws



Illustrative scene. Model's views of A and B differ by 3.67 / 255 RMS per pixel.

Before the lift, the pixels barely differ, but the physical state (contact) already does. Self-prediction from pixels has little signal to separate A from B.

Idea

1 Alignment
Simulators give time-aligned **camera** and **physical-state** trajectories at training time.

2 Cross prediction
One shared predictor forecasts **both** futures from **either** view, so the visual latent must carry contact and pose.

$$\mathcal{L}_{\text{JEPA-}x} = \sum_{m,n \in \{o,s\}} \left\| g_\psi(Z_t^m, a_t) - z_{t+1}^n \right\|_2^2 \; + \; \beta \left[\Omega(z^o) + \Omega(z^s) \right]$$

$m = n$: within-modal (keeps LeWM self-prediction) · $m \neq n$: cross-modal · Ω: SIGReg isotropy regularizer, $\beta = 0.09$

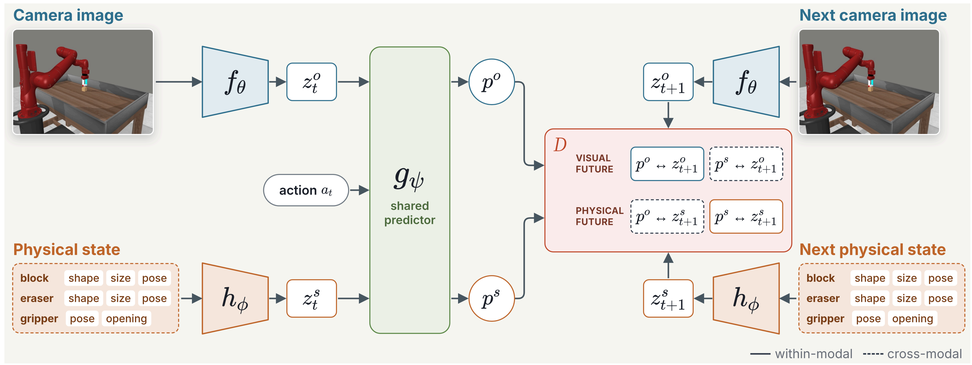
3 Visual-only deployment
The physical branch is removed at test time: pixels only, same planner as the baseline.

Takeaways

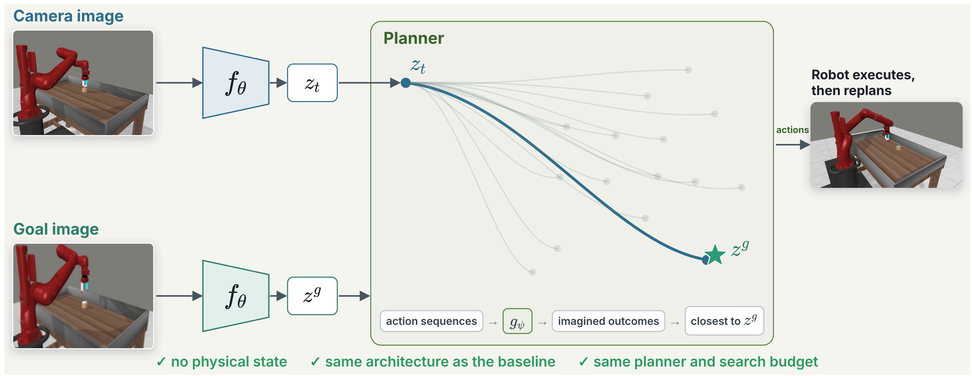
- Cross-modal prediction makes visual latents **more forecastable**, not just more decodable.
- Forecastable latents give **better planning from pixels**: 53.6 → 78.2% control.
- Physics is used **only in training**: same encoder, predictor and planner at test time.

2 Method

Training with paired trajectories



Deployment from pixels only



Planner: z-CEM over a 32-d action-VAE latent (5 macro-steps × 5 actions, 300 candidates, 30 iters), executes 25 actions, then replans. Same budget as the visual-only baseline LeWM (Maes et al., 2026).

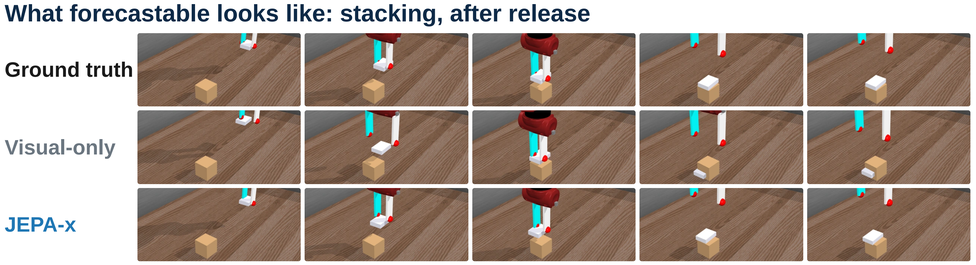
Decodable ≠ forecastable: cross-prediction is what improves control

Training objective	Decodable ↑ object-position R^2	Forecastable ↓ drift, new predictor	Useful ↑ control success %
Visual-only LeWM self-prediction	0.978	0.361	53.6
Regress + state regression	**0.991**	0.373	53.7
Cross-only separate predictors	–	0.101	73.0
JEPA-x shared predictor	**0.978**	**0.104**	**78.2**

Most decodable, no better control: regression makes position easiest to decode, yet the latent is no more forecastable. Cross-modal prediction is the main contributor; JEPA-x and Cross-only reach similar low drift.

Mean of 3 seeds. Distill: R^2 0.991, drift 0.516, control 57.7.

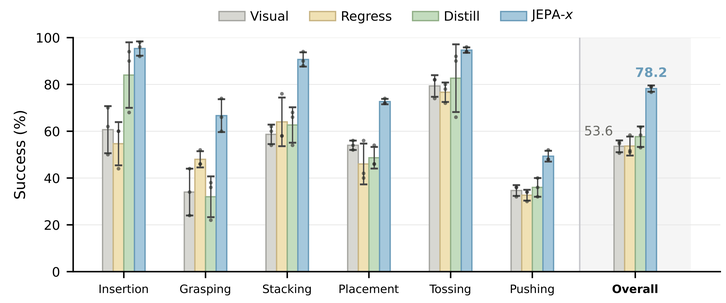
What forecastable looks like: stacking, after release



Decoded forecasts $t_0 \rightarrow t_4$ under identical actions (paper Fig. 5), not control executions. Visual-only predicts the eraser sliding off; JEPA-x keeps it supported.

3 Main Results

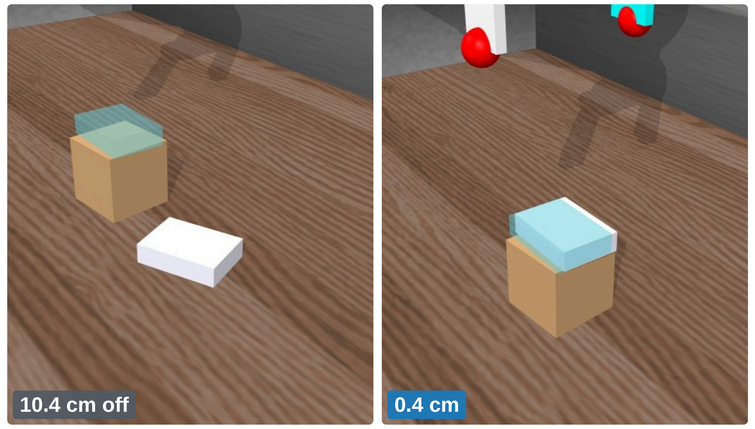
Control success per task family



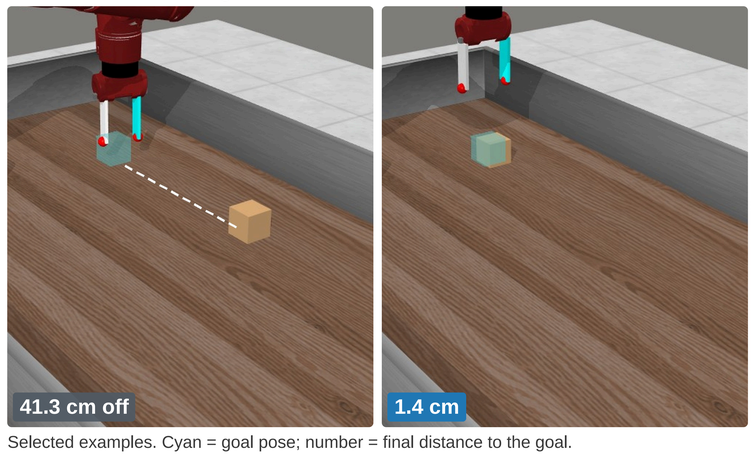
Same start, same goal

Visual-only | **JEPA-x**

Stacking eraser on block



Placement block to target



Selected examples. Cyan = goal pose; number = final distance to the goal.

Paper
arxiv.org/abs/2608.24044

Website
wkh923.github.io/jepa-x